

Supplementary Materials: Mathematical Derivations and Lean 4 Formalizations

Overview

This supplementary document provides the formal mathematical framework underlying the core claims presented in the main manuscript. The theorems below have been formalized and verified using the Lean 4 interactive theorem prover. Formalization ensures that the transitions from foundational physical primitives (symmetries, topology, and thermodynamics) to the complex structural properties of consciousness (boundaries, unity, and reflexivity) are stated precisely rather than metaphorically. It does not by itself make them physically true. The development now declares *no* axioms — `#print axioms` reports only Lean’s three foundational ones — but the physical postulates have not vanished; they have moved into class fields that each instance must discharge, and into the hypotheses of individual theorems. Five axioms in earlier drafts were removed after three were found to be logically inconsistent; see the soundness subsection of the main text. Where the Lean statement is weaker than the informal theorem above it, we say so explicitly in the implementation note. The complete Lean 4 source code is publicly available at https://github.com/bobaseb/physical_primitives_of_mind, and web-accessible documentation is available at https://bobaseb.github.io/physical_primitives_of_mind/.

1 Symmetry Breaking and the Emergence of Boundaries

The first step in our deductive chain establishes that a perfectly symmetric universe cannot support agency. Any distinguishable physical system requires a topological defect.

Let $\mathcal{M}$ be a pseudo-Riemannian manifold representing spacetime, and let $\phi : \mathcal{M} \rightarrow \mathcal{V}$ be a physical field mapping to a vacuum manifold $\mathcal{V}$ (the space of energy minima). Let G be a continuous symmetry group acting on $\mathcal{V}$.

Theorem 1 (Topological Defects from Broken Symmetry): If the continuous symmetry G is spontaneously broken such that $\mathcal{V} \cong G/H$ (where H is the unbroken subgroup), and the fundamental group $\pi_1(\mathcal{V}) \neq 0$, then field configurations ϕ defined on a spatial boundary $S^1 \subset \mathcal{M}$ cannot necessarily be smoothly deformed to a constant vacuum state.

Lean 4 Implementation (Phase1_Primitives.lean): We formalize this by defining the set of degenerate energy minima `DynamicalVacuum` and proving `boundary_defect_forces_interior_vacuum_break`: if a boundary field configuration ϕ_∂ is not null-homotopic, then it admits no continuous extension over a contractible interior. This geometric obstruction forms the fundamental physical boundary separating an “inside” from an “outside.” Note the direction: Lean establishes that a defect *obstructs extension*, not that a defect must exist. The companion lemma `contractible_interior_forces_trivial_boundary` (renamed from `defect_inevitability`, whose old name asserted the opposite of its statement) proves the contrapositive. Separately, `spontaneous_symmetry_breaking` shows a global energy minimiser lies in the vacuum manifold almost everywhere; this is now axiom-free, the pointwise step being `pointwise_vacuum_of_global_min`. `Examples.lean` §11 witnesses that result on the double well $V(v) = (v^2 - 1)^2$ over a two-point spacetime carrying a Dirac probability measure. Two features of the witness are deliberate. Its minimisation hypothesis is *proved* rather than assumed — everywhere else in the development the existence of a global minimiser is a hypothesis, since no compactness or direct-method argument is formalized. And the almost-everywhere in the conclusion is shown necessary: the minimising field sits at the vacuum on the support and at 0 off it, so the pointwise version of the statement is false of the very field the theorem applies to. The same section instantiates the $\mathbb{Z}_2$ action $v \mapsto \pm v$ and discharges the energy invariance, giving the first instance of `SymmetryInvariantAction`; the vacuum manifold $\{-1, 1\}$ is proved degenerate and the two minima exchanged by the symmetry, so a minimiser and its image have equal energy and differ. No conserved quantity is constructed from any of this, and there is still no Noether theorem.

2 The Thermodynamic Constraint: Landauer’s Principle

Once a boundary is established, it is subjected to external perturbations. Because the internal phase space of any localized physical system is finite, the system must overwrite internal states to adapt.

Let the global phase space $\Omega = \Omega_{sys} \times \Omega_{bath}$ undergo globally reversible Hamiltonian dynamics, modeled as an injective mapping $f : \Omega \rightarrow \Omega$.

Theorem 2 (Thermodynamic Erasure): If the local transition map on the system $\pi_{sys} \circ f$ is not injective (i.e., multiple initial states map to the same final state, contracting the local phase space), then by the conservation of global phase space volume (Liouville’s theorem), the bath phase space must expand, resulting in an irreversible transfer of heat to the environment: $\Delta Q \geq k_B T \ln 2$.

Lean 4 Implementation (Phase3.CombinatorialThermodynamics.lean): We explicitly decouple logical erasure from physical erasure. Lean verifies that any contraction in the *physical* state space ($\neg \text{Function.Injective}(t_{sys})$) under globally reversible dynamics mathematically necessitates heat dissipation. The combinatorial core, `landauer_from_reversibility` and `second_law`, is fully derived. Converting it into a statement about heat requires the postulate $\Delta Q = T \Delta S_{bath}$, which is carried as the class field `StatisticalMechanics.heat_eq`; `Examples.lean` exhibits a one-bit erasure system with an explicit bath that discharges it, so the postulate is demonstrably satisfiable. The boundary is thus forced to operate as a dissipative structure.

3 Structural Resonance and Least Action

To avoid catastrophic dissipation, the dissipative structure evolves to minimize entropy production. Let $P(ext)$ be the probability measure of external environmental perturbations, and $Q(int)$ be the internal transition probabilities of the system.

The total entropy production σ is bounded from below by the Kullback-Leibler divergence:

$$\sigma \geq \frac{1}{\Delta t} D_{KL}(P(ext) || Q(int)) \quad (1)$$

Theorem 3 (Structural Resonance): By the Principle of Least Action, the system undergoes gradient descent on σ , forcing $\lim_{t \rightarrow \infty} D_{KL}(P || Q) = 0$.

This conceptual mapping suggests that “prediction error minimization” is fundamentally a physical thermodynamic attractor. The information-theoretic KL bound is formalized in `Phase3.KLBound.lean` as the class field `StructuralResonance.kl_bound` — an irreducible physical postulate attached to a specific system rather than asserted globally, the earlier `kl_bound_axiom` formulation having been inconsistent. Gibbs’ inequality (KL non-negativity) is proved as a genuine theorem from $\log x \leq x - 1$; `structural_resonance_bound` restates the axiom in the form $D_{KL} \leq \Delta t \cdot \sigma$, and `discrete_entropy_rate_nonneg` combines the two to derive $\sigma \geq 0$. The gradient-flow half of Theorem 3 is now formalized against the entropy production functional itself: `Phase8.ContinuousField.lean` §7 derives the Fréchet derivative of σ with respect to the coupling kernel, identifies the differentiated functional with `entropy_production_rate` on finite substrates carrying counting measure, and concludes in `structural_resonance_decreases_entropy_production` that σ is non-increasing along the resulting flow; `Examples.lean` §5 exhibits a two-site substrate on which σ decays as e^{-2t} . What remains unformalized is the limit itself, $\lim_{t \rightarrow \infty} D_{KL} = 0$: monotone descent plus the bound $D_{KL} \leq \Delta t \sigma$ does not by itself force the limit to be zero, which would need a coercivity or Łojasiewicz-type estimate we do not have. The link is also restricted to finite substrates and to a fixed phase field, though the first restriction is narrower than the file previously claimed. `Phase8.ContinuousField.lean` §8 records the correction: σ is a quadratic functional of an affine image of the kernel, so `hasFDerivAt_quadratic_of_affine` differentiates it by the chain rule with no limiting argument, and `hasFDerivAt_sigmaOfKernel_of_operator` recovers §7’s gradient from that abstract lemma. The continuum case is consequently blocked on exhibiting the drift map as a bounded operator between L^2 spaces — a Cauchy–Schwarz estimate — rather than on differentiation under the integral sign. A separate result in the same file concerns the target of the descent rather than the descent itself, and it is now unconditional on the class of kernels the framework uses. `phase_locked_achieves_minimum_entropy` proved that the phase-locked field minimizes σ only among competitors sharing its spatial mean drift, which made the content Jensen’s inequality alone; the note recorded that removing the restriction would mean modelling how the mean drift varies with the competitor. It does not. For a symmetric kernel the integrand $K(x, y) \sin(\theta_y - \theta_x)$ is antisymmetric in its two sites, so by Fubini its double integral equals its own negative and vanishes (`coupling_integral_eq_zero`); the total drift is consequently the same for every phase field (`total_drift_eq_symm`), the phase-locked configuration fixes its value,

and the mean-matching hypothesis is a theorem (`mean_drift_of_symm`) rather than an assumption. `phase_locked_minimizes_entropy_of_symm` is the resulting statement, quantified over all competitor fields. Symmetry is imposed as a hypothesis rather than a field of `ContinuousNeuralField`, since the results quantify over instances, and it is load-bearing: `Examples.lean` §16 builds a one-way coupling on the two-site substrate whose phase-locked field is strictly beaten (`asymSys_locked_not_minimal`, $\sigma = 1/2$ against 1), with `asymSys_mean_drift_fails` isolating the mechanism — there the total drift is not conserved, so the variance bound has nothing to stand on. On the witness side, `Examples.lean` §2 now carries two instances rather than one. The original is perfectly resonant, $D_{KL} = 0$, and so discharges the postulate trivially. The second (`boolDetuned`) is the one-bit eraser of §1 driven by a point-mass environment against uniform internal statistics: there $D_{KL} = \log 2 > 0$ and $\sigma = \log 2 > 0$, and `boolDetuned.tight` proves the postulate holds with *equality*. The bound is therefore attained and cannot be strengthened to a strict inequality.

4 Macroscopic Scaling and Kuramoto Phase Synchronization

As the system scales, it forms a continuous macroscopic electrodynamic field. We model the local electromagnetic phases θ_i as coupled non-linear oscillators via the Kuramoto model.

For a system with coupling strength K and thermodynamic noise D , the dynamic evolution is:

$$\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i) + \eta_i(t) \quad (2)$$

Theorem 4 (Critical Phase Transition): When the integrated spatial coupling K exceeds the thermodynamic noise threshold $K_c = 2D$, the oscillators undergo a sudden phase transition into a synchronized, phase-locked state ($r > 0$).

Lean 4 Implementation (Phase4.KuramotoDynamics.lean): We formalize the Lyapunov potential $V(\theta) = -\frac{1}{2} \sum_{i,j} A_{ij} \cos(\theta_j - \theta_i)$ of the coupled field and prove two directions: `phase_locked_minimizes_potential` (the phase-locked configuration attains the global minimum whenever all couplings are positive) and `potential_min_implies_phase_locked` (any minimizer is phase-locked). Note the scope: this is a statement about the *static* potential landscape and holds for all positive coupling matrices. The critical threshold $K_c = 2D$ does not appear in this file. It is formalized separately, in `Phase8.SelfConsistency.lean` (see below), as a statement about the mean-field self-consistency equation with the stationary density assumed; deriving that density from the noise-driven stochastic dynamics would require the spectral theory of the associated Fokker-Planck operator, which is not available in Mathlib. The threshold is also validated numerically (`simulations/kuramoto.py`).

Lean 4 Implementation (Phase8.ContinuousField.lean): What the continuous-field file does carry is the threshold value itself, `critical_coupling D = 2 * D`, and a predicate `exhibits_phase_transition` for a substrate lying above it. Two cautions. First, the predicate is not connected to `is_continuous_kuramoto_trajectory`, `entropy_production_rate`, or the order parameter of `Phase4`. What it is connected to is the mean-field self-consistency equation $r = I_1(Kr/D)/I_0(Kr/D)$, whose bifurcation at $K = 2D$ we formalize in both directions (see the next paragraph): `exhibits_phase_transition_coherent` shows that a substrate satisfying the predicate admits a solution with $0 < r \leq 1$, and `exhibits_phase_transition_unique_coherent` that it admits exactly one. So the threshold is no longer a bare stipulation, but it is a stipulation about a stationary density rather than about a trajectory. Second, the predicate now requires the substrate's `volume` to be a probability measure, and this hypothesis is load-bearing rather than decorative. It compares `mean_field_coupling`, the kernel averaged over both arguments, against $2D$; on a substrate of total mass m that average carries a factor m^2 , so without normalization the comparison tracks the substrate's size rather than its coupling and can be satisfied by any kernel at all. `Examples.lean` §9 exhibits a system a factor of two below threshold whose unnormalized double integral nonetheless clears $2D$. Under the hypothesis, `mean_field_coupling_const` gives that a constant kernel has strength exactly K and `exhibits_phase_transition_const_iff` reduces the predicate to $K > 2D$, witnessed in §9 on both sides of the threshold.

Lean 4 Implementation (Phase8.SelfConsistency.lean): The self-consistency equation behind $K_c = 2D$ is formalized as real analysis. With the von Mises stationary density $\rho(\theta) \propto e^{a \cos \theta}$, $a = Kr/D$, taken as given, `besselRatio` is $I_1(a)/I_0(a)$ written as a ratio of interval integrals, Mathlib having no modified Bessel functions. `vonMisesM.eq_mul_vonMisesS` is the integration-by-parts identity $\int_{-\pi}^{\pi} \cos \theta e^{a \cos \theta} = a \int_{-\pi}^{\pi} \sin^2 \theta e^{a \cos \theta}$, exact because $\sin(\pm\pi) = 0$ kills the boundary term. `vonMisesC2_nonneg` proves $I_2(a) \geq 0$ by folding $[-\pi, \pi]$ onto $[0, \pi/4]$ through the reflections $\theta \mapsto \pi - \theta$ and $\theta \mapsto \pi/2 - \theta$, where

the integrand becomes a product of two manifestly non-negative factors; `vonMisesS.le_half_vonMisesZ` restates it as $\mathbb{E}_a[\sin^2 \theta] \leq 1/2$ and `besselRatio.le_half_self` as $R(a) \leq a/2$. That factor of $1/2$ is exactly what separates a threshold at D from one at $2D$; `subcritical_fixed_point_eq_zero` is the crude form obtained from $\sin^2 \leq 1$ and is kept to make the point visible. The subcritical theorem is `subcritical_fixed_point_eq_zero'`: for $0 \leq K < \text{critical_coupling } D$ and $D > 0$, the only $r \geq 0$ with $r = R(K, r)$ is $r = 0$. The supercritical theorem is `supercritical_fixed_point_exists`: for $K > \text{critical_coupling } D$ there is an r with $0 < r \leq 1$ and $r = R(K, r)$. Its proof needs no series expansion. Factoring $R(a) = a E(a)$ with $E(a) = \mathbb{E}_a[\sin^2 \theta]$ — the factorization the integration by parts supplies — it suffices that E is continuous with $E(0) = 1/2$: since $D/K < 1/2$, the map overshoots for small $r > 0$, while `besselRatio.le_one` (R is a mean of $\cos \theta$) makes it undershoot at $r = 1$, and `intermediate_value.Icc'` supplies the crossing. Continuity of E is Mathlib's continuity of a parametric interval integral, applied to the jointly continuous integrand. `critical_coupling_is_threshold` states the two halves together, and §9 of the file discharges the hypotheses of each at concrete values ($K = 3$, $D = 1$ above threshold; $K = 3$, $D = 2$ below it), so neither is vacuous. Section 8 of the file identifies the equation with the physics it is named for. `vonMisesDensity` is the normalized weight, proved a probability density on $[-\pi, \pi]$ (`vonMisesDensity_integral_eq_one`); `circularOrderParameter` is the average of $e^{i\theta}$ against a density, the continuum analogue of `order_parameter_complex` of Phase4; `circularOrderParameter_vonMises` proves that average equals $R(a)$, real because `vonMises_mean_sin` gives $\mathbb{E}_a[\sin \theta] = 0$; and `fixedPoint_iff_selfReproducing` states the equivalence: r solves $r = R(K, r)$ exactly when the density it induces has order parameter r . `incoherent_density_uniform` identifies the solution $r = 0$ with the uniform density, and `supercritical_coherent_density` restates the existence theorem as the existence of a stationary density with positive order parameter. Three results added since sharpen the threshold. `vonMisesC2_pos` upgrades the fold argument to a strict inequality for $a > 0$ — `foldB` is continuous, non-negative on $[0, \pi/4]$ and strictly positive on its interior, so `intervalIntegral_pos_of_pos_on` applies — giving $\mathbb{E}_a[\sin^2 \theta] < 1/2$ and $R(a) < a/2$ for every $a > 0$. That closes the threshold case: `fixed_point_eq_zero_of_le_critical` covers all $K \leq 2D$, so the incoherent and coherent regimes now exhaust $K \geq 0$ with no gap, and $K = 2D$ falls on the incoherent side. `coherent_iff_sRatio_eq` reformulates the coherent branch: for $r \neq 0$, solving $r = R(K, r)$ is equivalent to $E(Kr/D) = D/K$, so the branch is a level set of E . `coherent_branch_continuous_at_threshold` then proves that for every $\varepsilon > 0$ there is a $\delta > 0$ such that on $(2D, 2D + \delta)$ every coherent solution satisfies $r < \varepsilon$ — the branch emerges continuously from zero, excluding a discontinuous jump. Its proof needs no monotonicity of E : on a compact interval bounded away from the origin the extreme value theorem gives a maximum strictly below $1/2$, and beyond that the crude bound $E(a) \leq 1/a$ suffices. Section 7 of the file closes uniqueness, which the paragraph above reduced to the injectivity of E on $(0, \infty)$, and closes it in the stronger form `vonMisesSRatio_strictAntiOn`: E is strictly decreasing on $[0, \infty)$. The expected route — differentiate the exponential family, obtain $E'(a) = -\text{Cov}_a(\cos^2 \theta, \cos \theta)$, and show that covariance positive — is not the route taken, and it was the harder one: it needs differentiation under the integral sign, and the covariance is not sign-definite pointwise, since the symmetrised form $\frac{1}{2} \mathbb{E}[(X - X')^2(X + X')]$ with $X = \cos \theta$ has an integrand changing sign with $X + X'$. Instead the target is kept as the two-point inequality $C_2(a)Z(b) < C_2(b)Z(a)$ for $a < b$, and two moves settle it. *Fold first*: `vonMisesZ.eq.foldW` and `vonMisesC2.eq.foldA` carry both integrals onto $[0, \pi/2]$ by the two reflections already used for $I_2 \geq 0$, replacing $e^{a \cos \theta}$ by $W(a, \theta) = 2 \cosh(a \cos \theta)$ and confining $\cos \theta$ to $[0, 1]$; there `crossIntegrand_nonneg` shows $\cos 2x - \cos 2y$ and $W(b, x)W(a, y) - W(a, x)W(b, y)$ have the same sign, both being governed by $\cos x$ against $\cos y$. This is the same $X \mapsto -X$ symmetrisation the covariance form needed, performed once at the start rather than inside a double integral, and its analytic content is one identity: `cosh_mul_cosh_cross`, i.e. $\cosh(ap) \cosh(bq) \leq \cosh(bp) \cosh(aq)$ for $0 \leq a \leq b$, $0 \leq q \leq p$, from $2 \cosh X \cosh Y = \cosh(X+Y) + \cosh(X-Y)$ and $\cosh$ increasing in $|\cdot|$. *Then cross*: rather than integrating that inequality over a square, `foldA_mul_foldW.lt` takes α to be the mean of $\cos 2x$ under $W(a, \cdot)$ and $x_0 \in [0, \pi/2]$ a point where $\cos 2x_0 = \alpha$, supplied by `intermediate_value.Icc'`; evaluating at $y = x_0$ and integrating in x alone gives the conclusion, the $W(b, x_0)$ term vanishing by the choice of α . Strictness comes from `crossIntegrand_pos` on a subinterval missing x_0 . The same crossing argument run with $\cos \theta$ in place of $\sin^2 \theta$ needs no fold, since $R(a) = \mathbb{E}_a[\cos \theta]$ is the mean of the tilting variable itself, and gives `besselRatio_strictMono`: R is strictly increasing on all of $\mathbb{R}$. The consequences are `coherent_fixed_point_unique` (at most one positive solution), `supercritical_fixed_point_existsUnique` and `supercritical_solution_set` (the non-negative solutions are exactly $\{0\}$ at or below threshold and $\{0, r\}$ above it), and `coherent_branch_strictMono`, which uses both halves: E decreasing puts the concentrations of two couplings in the order $a_1 < a_2$ through $E(a_i) = D/K_i$, and R increasing carries that order to $r_1 < r_2$. Scope: the density is assumed, not derived from the Fokker–

Planck equation; E and R are shown strictly monotone but not differentiable, and no rate is proved; dynamical selection is not established, and could not be here, since no theorem in the file mentions a trajectory; and the order parameter reached is that of a density, not of a trajectory — no theorem relates `circularOrderParameter` to `order_parameter_complex`, which would be the mean-field limit. Every declaration in the file depends only on `propext`, `Classical.choice` and `Quot.sound`.

Lean 4 Implementation (Phase4RotatingFrame.lean): The potential above omits the natural-frequency term $-\sum_i \omega_i \theta_i$ that appears in the Lyapunov descent result `dV_dt_le_zero` of `Phase3.CombinatorialThermodynamics`. The omission is not cosmetic. With any $\omega_i \neq 0$ the full potential is unbounded below (`kuramoto_potential_unbounded_below`), so it has no minimum for a phase-locked state to attain, and the two results concerned different functionals. This file chains them by the standard rotating-frame reduction $\theta_i \mapsto \theta_i - \Omega t$. For a system of identical natural frequencies $\omega \equiv \Omega$: `is_kuramoto_trajectory_rotate` sends a trajectory to a trajectory of the zero-frequency system `sys.reduced` — the same system Theorem 5’s `ThermodynamicCover` works with; `kuramoto_potential_reduced` shows the two potentials coincide on it; `dynamic_potential_descent` transports the descent identity $\dot{V} = -\sum_i \dot{\theta}_i^2$ onto the potential above; and `rotating_frame_chain` runs this through the minimum characterization to phase-locking and $r^2 = 1$. Phase differences, phase-locking and the order-parameter magnitude are proved frame-invariant (`rotate_sub`, `is_phase_locked_rotate`, `order_parameter_r_sq_shift`), so no observable is lost. `Examples.lean` §7 runs the chain end to end on a two-oscillator system, discharging every hypothesis, and exhibits a system whose full potential has no minimum. Scope: the reduction is exact only for identical frequencies. A spread of frequencies leaves residual detunings $\omega_i - \Omega$ in the reduced system, to which the unboundedness result still applies, and phase-locking then depends on K_c , which this file does not treat — see the `Phase3.SelfConsistency.lean` note above for what is proved about the threshold, and note that it concerns a stationary density rather than the trajectories this file transports.

Trajectories that run. Descent along a trajectory is worth little if the only trajectory in the development starts at its own limit, which was the case until §7’s rigid rotation was joined by the results below. `kuramotoField_lipschitz` (`Phase3.CombinatorialThermodynamics.lean`) proves the Kuramoto vector field globally Lipschitz with constant $2\sum_{ij} |A_{ij}|$ — $\sin$ is 1-Lipschitz and the index set is finite, so there is no smallness caveat and the natural frequencies drop out. `is_kuramoto_trajectory_unique` turns this into Picard–Lindelöf uniqueness on all of $\mathbb{R}$: two solutions agreeing at one instant agree forever, in both time directions. Existence is now proved in general, and the gluing that Mathlib does not supply is written out. Mathlib’s `IsPicardLindelof` caps a local solution’s lifetime by the radius of the ball on which the field’s bounds hold, divided by the field’s sup norm there, and carries no global-in-time theorem; but `kuramotoField_norm_le` bounds this field by $\sum_i |\omega_i| + \sum_{ij} |A_{ij}|$ on the *whole* state space, so the cap can be defeated by enlarging the ball after choosing the window, and `kuramoto_exists_on_window` produces a solution on every $(t_0 - T, t_0 + T)$. Uniqueness on an interval makes the family coherent — solutions on nested windows agree where both are defined — so $\theta(t) := \alpha_{[t-t_0]}(t)$ is a solution on all of $\mathbb{R}$ (`is_kuramoto_trajectory_exists`), and `kuramoto_cauchy_problem` states existence and uniqueness together as a single `ExistsUnique`: the trajectory predicate is inhabited exactly once for each initial state, rather than only by the solutions we write down by hand. What existence does not supply is behaviour, and for that a concrete solution is still what we have. On two oscillators with unit coupling and zero natural frequency the phase difference obeys $\dot{\Delta} = -2\sin \Delta$, whose solution $\Delta(t) = 2\arctan(ce^{-2t})$ splits symmetrically into an exact trajectory of the full system (`Examples.lean` §15, `pairRelax_is_trajectory`). At $c = 1$ it starts a quarter turn out of phase (`pairRelax_not_locked_at_zero`); the cosine of the phase difference tends to 1, the order parameter tends to 1 through the two-oscillator identity $r^2 = (1 + \cos \Delta)/2$, and the dynamic potential tends to its global minimum. By the uniqueness result it is the only solution through its initial state. Two general statements about the long run are now proved as well, both for a zero-frequency system, which by `kuramoto_potential_unbounded_below` is the only case in which the potential is bounded below at all. `dynamic_potential_antitone` upgrades the instantaneous descent to monotonicity in t — `dynamic_potential_deriv_nonpos` constrains each instant separately and does not by itself say the trajectory settles — and with the uniform lower bound $-\frac{1}{2}\sum_{ij} |A_{ij}|$ monotone convergence gives `dynamic_potential_tendsto`: the potential converges along *every* trajectory. Integrating the Lyapunov identity gives the quantitative form, `dissipation_integral_eq`: energy dissipated over $[0, T]$ is exactly the potential dropped, so by `dissipation_integral_tendsto` the improper integral $\int_0^\infty \sum_i \dot{\theta}_i^2 dt$ converges, to the total drop. This needed the Lyapunov identity in `HasDerivAt` form rather than as a statement about `deriv`, which carries no differentiability information and cannot be fed to the fundamental theorem of calculus. `rotating_frame_dissipation` transports both to a uniform-frequency system. Finiteness of that integral is not the same as $\sum_i \dot{\theta}_i^2 \rightarrow 0$: an integrable function can spike forever on ever narrower intervals. The classical bridge is Barbalat’s lemma,

which Mathlib does not have, so it is proved here as `tendsto_zero_of_lipschitz_of_integral_le`: a non-negative Lipschitz function with $\int_0^T g$ bounded above tends to zero, because a spike of height ε drags a window of width $\varepsilon/2K$ up with it and only finitely many such windows fit under a finite integral. Its hypothesis is then checked for the dissipation rate without differentiating it: the velocities are bounded by $\sum_i |\omega_i| + \sum_{ij} |A_{ij}|$, the field is Lipschitz in the configuration, and the configuration is Lipschitz in time because its derivative *is* the bounded field (`trajectory_dist_le`, by the mean value theorem, with no invariant region needed since the bound holds on the whole state space). The conclusion is `velocity_sq_tendsto_zero`: along *every* trajectory of a zero-frequency system, $\sum_i \dot{\theta}_i^2 \rightarrow 0$; equivalently each $\dot{\theta}_i \rightarrow 0$ and the vector field along the trajectory tends to 0 in the state space. This is the first asymptotic statement here about an arbitrary trajectory rather than about a witness, and `Examples.lean` §15 fires it on the relaxing pair, whose velocity starts at -1 and is therefore genuinely decaying rather than identically zero. It does not locate the limit. Locating it we had recorded as blocked on the LaSalle invariance principle, absent from Mathlib, and on the state space $V \rightarrow \mathbb{R}$ carrying no compact invariant set; both parts of that were wrong. LaSalle is what one needs when a decreasing functional is all one has, and here the flow is a gradient flow of the dynamic potential — ∇_i of `kuramoto_potential_dynamic` is exactly $-\text{kuramoto_velocity}$ at zero frequencies, which is why `dynamic_potential_hasDerivAt` gives $\dot{V} = -\sum_i \dot{\theta}_i^2$ as an identity — whose minimisers are already characterised. `lojasiewicz_estimate` adds the missing ingredient: on the region where every pairwise phase difference lies within a quarter turn, $2a^2W \leq \left(\sum_{ij} A_{ij}\right) \sum_i \dot{\theta}_i^2$, where W is the excess of the potential over its minimum and $a > 0$ is a lower bound for the coupling. Its proof uses no spectral gap and no convexity, only two extremal indices: at the site m carrying the largest phase every coupling term has $\theta_j - \theta_m \in [-D, 0]$ with $D \leq \pi/2$ the spread, so every term is non-positive, and the term from the site carrying the smallest phase is at most $-a \sin D$ (`velocity_max_le`); against $W \leq \frac{1}{2} \left(\sum_{ij} A_{ij}\right) (1 - \cos D)$ the estimate follows from $\sin^2 D = (1 - \cos D)(1 + \cos D)$ and $\cos D \geq 0$, which is the only place the quarter-turn hypothesis is used. The compactness objection is met without any invariant region put in by hand. Because all terms of the excess are non-negative, a single pair satisfies $\frac{1}{2}a(1 - \cos(\theta_j - \theta_i)) \leq W$, so an initial excess below $a/2$ — which `dynamic_potential_antitone` preserves — keeps every pairwise cosine strictly positive for all time. A difference escaping $[-\pi/2, \pi/2]$ would have to cross $\pi/2$ in absolute value by the intermediate value theorem, and there the cosine vanishes; so the quarter-turn region is forward invariant (`spread_le_of_small`). This is branch-pinning by continuity on a connected half-line, and it replaces the quotient by the phase-shift symmetry we had thought necessary. The differential inequality $\dot{W} \leq -cW$ with $c = 2a^2 / \sum_{ij} A_{ij}$ then integrates, in the elementary form “ $W(t)e^{ct}$ has non-positive derivative”, to $W(t) \leq W(0)e^{-ct}$ (`excess_decay`). The rate transfers to the velocity through $|\sin d| \leq \sqrt{2(1 - \cos d)}$, giving $|\dot{\theta}_i(t)| \leq Be^{-ct/2}$ (`velocity_abs_le_exp`), which is integrable on $(0, \infty)$; since $\theta_i(t) = \theta_i(0) + \int_0^t \dot{\theta}_i$, the trajectory converges (`phase_tendsto`), and $W \rightarrow 0$ forces the limit to be phase-locked, hence a global minimiser with $r^2 = 1$ (`kuramoto_tendsto_global_minimum`). `rotating_frame_tendsto_global_minimum` transports it to uniform frequencies, where what converges is the relative configuration. The two hypotheses are on the initial data and neither can be removed: splay and twisted configurations are equilibria too, so “every trajectory reaches the phase-locked state” is false, and what is proved is the arc condition version — which the two-oscillator witness satisfies for every c , since $2 \arctan$ lands in $(-\pi, \pi)$. `Examples.lean` §17 runs the theorem on three sites, where the method of §15 is unavailable because the system has no closed-form solution: the trajectory is produced by `is_kuramoto_trajectory_exists` from the initial data $(0, 0, \frac{1}{2})$, which is not phase-locked, and its limit by the convergence theorem as $\theta(0) + \int_0^\infty \dot{\theta}$; the numerical content of checking the hypotheses is $\cos \frac{1}{2} > \frac{3}{4}$, supplied by `Real.cos_bound`.

From the continuous field to the discrete model (`Phase2_SimplicialBridge.lean`, `Phase2_MeshConvergence.lean`): The coupling matrix A_{ij} above is not postulated. A triangulation of the substrate assigns to each edge (u, v) a region of the manifold, and the coupling weight is the Lebesgue–Bochner integral of the scalar magnitude of the stress-energy tensor over that region (`edge_weight`); symmetry and non-negativity of A are then proved rather than assumed. The bridge back is mesh refinement: as the regions shrink, the discrete energy $\frac{1}{2} \sum_{u,v} \mu(R_{uv}) f(x_u)$ should converge to the continuous energy $\int_S f$. This was previously stated as a conjecture and was in fact false as stated — nothing forced the regions to cover the manifold, and empty regions have diameter zero, so the everywhere-empty triangulation refuted it. It is now a theorem (`mesh_refinement_convergence`), quantified over a sequence of triangulations of a region S whose regions are measurable, pairwise disjoint across distinct edges, covering, and anchored to the embedded vertices, with mesh size tending to zero. The fixed-mesh error is bounded by $\varepsilon \mu(S)$, with ε a modulus of continuity of the integrand, and the symmetric double sum is proved equal to the midpoint rule over

unordered edges. Two limitations are worth stating. The theorem gives convergence, not the $O(1/N^2)$ rate reported by `simulations/mesh_refinement.py`. And `TriangulatedManifold` itself is bare data: it pairs a complex, an embedding and a family of edge regions without relating them, so the geometric conditions live in a separate predicate `IsRegularTriangulation`. Because those conditions are in tension — cells disjoint yet covering, every vertex anchored to each of its edges — `Examples.lean` §6 exhibits an instance: the uniform partition of $[0, 1]$ into N half-open cells, on which the convergence theorem is instantiated and the approximations are checked to genuinely change with N .

The discretization structure `DiscreteThermodynamics` now *carries* that predicate as a field, which is what makes its edge weights integrals over a geometrically constrained family rather than over arbitrary sets, and three statements follow that were unavailable before. `weight_self`: a vertex has no edge to itself, so the coupling matrix has zero diagonal. `face_of_weight_ne_zero`: a nonzero weight $w(u, v)$ forces $\{u, v\}$ to be an actual face of the complex, so the discretization cannot invent couplings between vertices the triangulation does not join. And `total_weight_eq_setIntegral`: $\frac{1}{2} \sum_u \sum_v w(u, v) = \int_S |T|$, i.e. the total coupling strength is exactly the scalar stress-energy of the region covered — nothing counted twice (disjointness), nothing dropped (covering). That last identity is the precise sense in which the coupling matrix is derived from the continuum rather than posited, and it is exactly what the everywhere-empty triangulation violates. `Examples.lean` §6 witnesses the strengthened structure on the same grid: each consecutive edge carries weight $1/N$, and the weights sum to 1 for every N . The witness is one-dimensional and its stress-energy tensor is constant, so it establishes satisfiability, not a physically interesting discretization. One field of the structure is deliberately left free and is now marked `[MODELLING]: scalar_magnitude`, the scalar read off the stress-energy tensor, constrained only to be non-negative. A $(0, 2)$ tensor has no canonical scalar — the trace, an observer’s energy density and the tensor norm are different functions, and the structure carries neither a metric nor a frame — so imposing one would pin a choice rather than derive it, and the three statements above hold for every non-negative choice, so it would prove nothing new. That the choice is nonetheless a physical input is stated as a theorem rather than left as a remark: §6 exhibits a second instance over the same triangulation, tensor and regions, differing only in that field, for which the partition identity again holds while every edge weight differs (`gridThermo_magnitude_matters`). The remaining constraint, non-negativity, is the discrete shadow of an energy condition and is what makes the induced couplings a legal Kuramoto coupling.

5 Global Sections and the Unity of Consciousness

Phase synchronization alone is mathematically insufficient for conscious unity; the synchronized local fields must be topologically glued together.

Theorem 5 (Sheaf-Theoretic Unity): Given a presheaf of probability measures $\mathcal{F}$ over a continuous biological substrate X , if local invariant measures $P_U \in \mathcal{F}(U)$ agree on all spatial intersections $U \cap V$ (guaranteed by the Kuramoto phase-locking), they uniquely glue into a singular Global Section $P_X \in \mathcal{F}(X)$.

Lean 4 Implementation (Phase5_GlobalSection.lean): We apply sheaf theory to prove that phase-locked local fields fuse into a single global section, via `global_section_from_thermodynamics`. One caveat and one discharged obligation. The caveat: phase-locking is not derived from dynamics here, it is the class field `ThermodynamicCover.thermodynamic_equilibrium`, so the theorem reads “given a cover already at the potential minimum, the local sections glue uniquely.” The discharged obligation: an instance of `ThermodynamicCover` now exists (`Examples.lean`, §4), so the theorem is no longer conditional on a structure of unknown realizability. The witness is a three-site substrate with the discrete topology and its Borel σ -algebra, covered by two overlapping open patches meeting in one shared site; local sections are germ families of the phase-indexed measure $\mu_\theta = (1 + \cos \theta)^+ \delta_{\text{mid}}$ in the sheafification of the probability presheaf. Two details matter for non-vacuity: μ_θ is provably non-constant in θ (`phaseMeasure_not_const`), so the periodicity obligation this witness carries is a real 2π -periodicity condition rather than a triviality; and `thermodynamic_equilibrium` is discharged from `phase_locked_minimizes_potential` rather than assumed. We therefore claim a formalized, non-vacuous gluing mechanism — not a solved binding problem, since the witness is a toy substrate whose phases are locked by construction. Three results added since then sharpen what the theorem does. First, `potential_min_iff_phase_locked` (`Phase4_KuramotoDynamics.lean`) characterizes the minimizers of the reduced potential in both directions — they are exactly the phase-locked configurations — from which `ThermodynamicCover.phase_locked` follows for *every* instance. Second, `Examples.lean` §13 exhibits the strongest witness the original structure allowed: the phase field is not the constant function (`twistPhase_not_const`, the second patch one full turn ahead), the invariant measure spreads three

different masses over the three sites (`richDensity.zero.not.uniform`) instead of §4’s single point mass, and the patch-local densities are read off the sections themselves through `densityOn`, so the agreement on the shared site is an equation between two separately computed numbers rather than a syntactic identity. It cannot be improved on in the direction one would want: by `ThermodynamicCover.phase.locked` no instance can carry phases that differ by anything a measure is allowed to see.

Third, and this one changed the theorem rather than the witnesses. As stated until 2026-08-30, Theorem 5 in Lean was uniqueness of the global state given the local data, not emergence of a global state from it: the class field `phase.invariant.measure` handed each instance a section over $\mathbb{T}$, and `sync.to.section.eq` declared each local section to be a restriction of it, so the existence half was assumed and the sheaf condition only excluded competitors. Both fields have been removed. `LocalSectionSynchronization` now constrains its local data only by `section.agrees.of.phase.eq` — patches at a common phase agree where they overlap — in which every symbol ranges over one patch or one overlap. The sheaf property is isolated as `probability.glue.unique`, a lemma with no physics in it, which takes a compatible family to the unique section over $\mathbb{T}$ restricting to it; `ThermodynamicCover.invariantMeasure` is defined as that section, and the deleted field reappears as a theorem about it, `ThermodynamicCover.sync.to.section.eq`. The old shape survives as the constructor `LocalSectionSynchronization.ofInvariantMeasure`, which is what §4 and §13 are now built through — proof that the weakened class is genuinely weaker, and the reason those two witnesses still glue to the measure they started from. `Examples.lean` §14 is built without it: the two patches carry independently chosen mass profiles (2, 1, 7) and (5, 1, 3), agreeing only at the shared site where both put mass 1, and the section the gluing returns has profile (2, 1, 3). It is neither profile, and neither profile restricts correctly to both patches (`leftW.glues.nothing`, `rightW.glues.nothing`), so the global state was produced by the theorem and not supplied to it. One caveat on that witness has since been removed: its patch-local sections were originally *defined* by restricting a measure on the whole substrate, because the germ-measure dictionary of §10 was built at $\mathbb{T}$ only. That dictionary is now stated at an arbitrary open — `densityOn` reads a section over any U , `sectionOfMassOn` builds one there, and `massEquivOn` is the resulting bijection with the mass profiles on U — so each patch’s section is constructed over its own patch and no measure on the whole substrate appears in the instance. `glued.section.eq.restrict` records that the object is unchanged, so the construction and not the content is what moved. What remains assumed is the overlap agreement itself, and `thermodynamic.equilibrium`: no dynamics is run into the minimum.

6 Reflexive Topology and the Self

To minimize internal thermodynamic friction, the unified Global Section must model its own internal state changes.

Theorem 6 (Auto-Resonance): The system minimizes total entropy production only if the internal state mapping $Q(int)$ contains a low-dimensional reflexive projection of its own boundary $\partial\mathcal{M}$.

Lean 4 Implementation (Phase6ReflexiveTopology.lean): We formalize the reflexive projection as a contraction on a complete metric space and apply the Banach fixed-point theorem to obtain a unique fixed point — the “Self” as a topological feature rather than a metaphysical addition. The entropy production functional does not appear in this file: the identification of the Banach fixed point with the minimizer of σ is an interpretive step made in the main text, and the contraction constant (1/2) is assumed rather than derived from field dynamics. Until recently no instance of `ReflexiveBoundary` or `PredictiveModel` existed anywhere in the development, and no metric on `GlobalSection` was ever constructed, so the theorem was conditional on structures of unknown realizability — the same failure mode an uninhabited class has. `Examples.lean` §10 removes that doubt on the three-site cortex of §4, and the metric it supplies is now built from the measures rather than imposed. Three stages. (i) The germ-measure dictionary: on a discrete substrate every point has a smallest open neighbourhood, so “mass carried at x ” is a well-defined map out of the stalk at x (constructed as the colimit map of a cocone whose cocone law is precisely that restriction preserves point masses); reading a section’s germ at each site of an open U gives a map to mass profiles on U , and `massEquivOn` proves it a bijection — injective because a germ on a discrete space is determined by its restriction to the point, surjective because every prescribed profile is realized by a finite measure on U . `massEquiv` is the case $U = \mathbb{T}$; the general case was added on 2026-08-30 and is what lets Derivation 5’s §14 witness build its patch-local sections over their own patches. Mathlib supplies nothing here: `TopCat.Presheaf.sheafify` has no adjunction and no `isIso.toSheafify`, so the inverse had to be built. (ii) The metric: the distance between two sections is the largest difference between the masses their measures put on a site, i.e. the uniform distance between their densities (equivalent to the total variation of the glued mea-

sures on a finite substrate, within a factor of the number of sites, but not equal to it). It is complete, proved by exhibiting the limit section of a Cauchy sequence of densities, and `exists_dist_lt_one` exhibits distinct sections at distance $1/2$. (iii) The Self: `relax` averages the current field with the baseline; `relax_dist` proves it contracts distances by exactly one half, `relax_not_const` that it is not constant, and `relax_fixed_unique` that its fixed point is unique. The earlier 0/1-metric witness is kept as `gsDiscreteMetric`, together with `contracting_implies_const`, which proves that under that metric every contraction is constant — the record of why the earlier witness said nothing. (iv) The avatar, added 2026-08-30. `ReflexiveBoundary.auto_resonance` is a function from the global state to a section over the avatar region, and until this pass nothing constrained it; since `reflexive_topology_implies_self` consumes only the predictive model, no theorem in the development mentioned the field at all, and a boundary whose avatar reads a constant satisfied every hypothesis and reached the same conclusion. Following the rule that data classes stay bare, the constraint is a predicate and not a new field: `IsRestrictionResonance rb` says `rb.auto_resonance s` is the presheaf restriction of s to the avatar region. Two theorems fix its status. `eq_of_avatar_eq`: if the avatar regions of a family of resonant boundaries cover the substrate, two global states with the same reading at every avatar are equal — the readings are local, and the global state is reconstructed from them by `probability_glue_unique`, the same sheaf condition Derivation 5 uses. `constResonance` and `self_of_constResonance`: from any boundary one builds another with the same region and the same predictive model whose avatar ignores its input; it is a legal `ReflexiveBoundary`, and the Self theorem applies to it verbatim with the same fixed point. `constResonance_not_isRestrictionResonance` shows it fails the predicate as soon as two global states differ on the avatar region. On the cortex both halves are exercised: `cortexReflexive` was already built from the restriction and discharges the predicate by `rfl`; `cortexReflexive_restrict_ne` checks the avatar region separates the silent field from the baseline, so the predicate is not vacuous there; `siteReflexive` gives one avatar per site, and since the singletons cover, `cortex_eq_of_avatar_eq` reconstructs the global state from three local readings. `cortexBlind` is the blinded boundary, with `cortexBlind_hasSelf` (the Self survives) and `cortexBlind_not_determined` (two distinct states with identical readings). We also record that Banach’s uniqueness half, which the original statement did not carry, is now available as `self_unique`. Three limitations remain and we state them. The map `relax` is a modelling choice: no field dynamics in the development produces it, and its contraction constant $1/2$ is built into its definition rather than derived from an entropy production rate. Nothing derives the resonance condition either: that a physical avatar performs the restriction is an assumption on an instance, in the same class as the overlap-agreement obligation of Derivation 5. And the Lévy–Prokhorov metric named in the file header is *not* what is constructed: on a discrete substrate the ε -thickening of a set is the set itself for $\varepsilon < 1$, so $\mu \mapsto \frac{1}{2}\mu + \frac{1}{2}\mu_0$ is not a Lévy–Prokhorov contraction there, and Mathlib carries no completeness result for that metric. The uniform distance between densities is the metric this substrate supports; we make no claim that it and Lévy–Prokhorov are equivalent.

7 Topological Rigidity of von Neumann Architectures

Finally, we compare the continuous coupling matrix $A(x, y)$ of biological fields with the discrete, rigid routing matrix G_{ij} of standard artificial hardware (e.g., GPUs).

Theorem 7 (Hardware Thermodynamics): Because a continuous parameter space contains global minima inaccessible to a discrete lattice under identical thermodynamic constraints, the dynamical potential of a biological neural field is strictly lower than that of rigid silicon hardware: $V(\text{continuous}) < V(\text{discrete})$.

Lean 4 Implementation (Phase7.HardwareComparison.lean): The lemma `exists_better_coupling_allocation`, wrapped as the theorem `continuous_beats_rigid_topology`, proves the reallocation step: given a coupling matrix satisfying `is_strictly_suboptimal` (positive weight on a less phase-correlated pair while a more correlated pair exists), there is another valid coupling with identical total resources and strictly greater total phase correlation.

Lean 4 Implementation (Phase7.Rigidity.lean): Two defects of the above are addressed here. (i) *Rigidity as a support constraint.* `RealizableIn S R A` says A is a valid coupling meeting resource budget R and vanishing on every pair the architecture has no wire for — weights tunable, wiring fixed. `totalCorrelation_le_of_realizable` proves no realizable coupling exceeds $R \cdot M$ when M bounds phase correlation over the wired pairs, and `exists_realizable_pair` attains it. `rigid_is_strictly_suboptimal` then *derives* `is_strictly_suboptimal` for every realizable coupling, whenever some pair (k, l) the wiring misses has correlation exceeding M : the hypothesis Theorem 7 previously assumed is now a consequence of the wiring pattern. `rigid_gap` gives the quantitative form, a gain of at least $R(\cos(\theta_l - \theta_k) - M) >$

0 at equal budget. `Examples.lean` §8 witnesses this on three sites wired only to the maximally anti-correlated pair, where the gap is exactly 2, and runs `exists.better.coupling_allocation` on it with no assumed hypothesis. (ii) *Continuity as a measure-theoretic distinction*. `fieldCorrelation` integrates a kernel against $\mu \otimes \mu$ as in Derivation 4. A finitely-sited architecture is a `Finset X`; `fieldCorrelation_sited_eq_zero` proves that on an atomless substrate such a kernel contributes exactly zero, for every phase field and every choice of weights, while `fieldCorrelation_patchKernel` gives $c\mu(U)^2$ for a patch. `sited_architecture_below_field_optimum` combines them; `Examples.lean` §8 instantiates it on $\mathbb{R}$ with a thousand units of weight at the origin against the unit interval.

Scope caveat. Result (i) is resource-matched and no longer question-begging, but it turns on fixed wiring support, not on continuity: an architecture wired to the best pair is not beaten. Result (ii) is a genuine type-level separation but is *not* resource-matched, since a measure-zero substrate carries zero resource in the $\mu \otimes \mu$ sense as well; what it establishes is that the continuum coupling energy cannot register a finitely-sited architecture at all, which is a claim about the model’s observable rather than about computational power. Neither result formalizes manifold structure or lattice geometry. Accordingly, the inference to “von Neumann architectures cannot experience unified consciousness” remains an informal argument in the main text, now with two verified steps rather than one.